A mini course in meta-analysis

Witold Więcek

February 2026

Meta-analysis

Meta-analysis: A statistical technique for combining results from multiple studies on a similar topic to obtain a quantitative summary estimate of the effect size.

Goal: a summary estimate of the effect size (that is more efficient than individual studies)

(what does "similar" mean? how to combine? why "more efficient"?)

Why meta-analysis

Expanding the goal into a somewhat aspirational list:

1. Provides a more comprehensive and precise estimate of the effect size by pooling data from multiple studies.
2. Increases generalizability of results.
3. Helps to resolve inconsistencies or contradictions in individual studies.
4. Allows for exploration of heterogeneity, subgroup analyses, cost-benefit analyses

(can you always be more precise and more generalisable?)

History

- Cochran (1953): averaging means (several unbiased estimators) via inverse-variance weighting
  - Building up from Cochran and Fisher ideas dating back to 1930s
- 1970-1990: RCTs as basis of clinical decision-making and "evidence-based" medicine (Archie Cochrane)
- Glass (1976): Primary and secondary research, meta-analysis
  - Coins "meta-analysis"
  - Many estimation methods established throughout 1980s
- Rubin (1981): Estimation in parallel experiments
  - Bayesian hierarchical model, "random effects" (also DerSimonian and Laird, 1986)
  - Much rapid advance only from 1990s due to computational advances allowing for MCMC
- Exponential growth in number of meta-analyses since 1990s
  - Cochrane Collaboration (1993) and Campbell Collaboration (2000): systematic reviews in health and beyond
  - Guidelines for conduct of meta-analyses also start in 1990s

What is and isn't meta-analysis

Basic meta-analysis model:

- pairs (b_i, se_i) such that $b_i \sim \mathcal{N}(\theta_i, se_i^2)$
- true effects $\theta_i \sim \mathcal{N}(\mu, \sigma^2)$
- what are μ and σ?

I will say "effects" but you could say "exchangeable measurements"

You can relax it in a myriad of ways:

- Non-normal errors
- Systematic differences between θ's (explained or unexplained heterogeneity)
- Analysing individual-level data
- Complex models (e.g. fuzzy RDD from 1000's of experiments!)

Conducting a meta-analysis

1. Define **research question** and objectives.
2. Conduct a systematic **literature review** to identify relevant studies.
3. **Extract data** from each study, including sample size, effect size estimates, and study characteristics.
4. Perform **statistical analyses** (may include preparation of individual studies before meta-analysing, to increase precision)
5. Assess **risk of bias** within studies and **publication bias**.
6. Interpret the results, predict future impacts.

You can ~ignore 2 and 5 if you are the designer of experiments!

Effect size

Quantity we decide to pool across studies.

(Again: not necessarily "effect"—can be repeated measurement)

Should be:

- Interpretable (what does the parameter measure?)
- Relevant (to the research question)
- Statistically comparable and reliable

Example measures:

- Means, proportions, correlations
- Mean differences, risk and odds ratios, IRRs, HRs, IVs

Internal and external validity

Quick reminder:

- Internal: within the studied sample?
- External: when considering other samples (groups, times, locations)?

Variation in effect sizes and true heterogeneity

Reported effects will always vary due to sampling error, even if $\theta_1 = \theta_2 = \ldots = \theta_K$

Heterogeneity: *additional* variability in true effect sizes:

$$\sigma \quad \text{in} \quad \theta_i \sim \mathcal{N}(\mu, \sigma^2)$$

- Can we explain it?
- Sources of heterogeneity: covariates, design, context
- Substantial heterogeneity poses questions of validity/generalisability (see next slide)
- In frequentist approaches it is typical to test if $\sigma = 0$

Critiques of meta-analysis

- **Cheating**: researcher degrees of freedom in (1) choice of studies, (2) building models
- **Selectivity**: Meta-analyses make it easy to ignore or underweight non-RCT evidence
- **"GIGO"** (garbage in, garbage out) issues:
  - Statistical power issues also apply to meta-analyses
  - Meta-analysis of low quality (small) studies can lead to overconfidence
  - Risk of giving extra credibility to poor research
- Institutional critiques:
  - John Ioannidis, 2016: 3% of meta-analyses are "decent and clinically useful" (speculative)
  - Richard Horton (editor in chief, the Lancet), 2019: "Imagine if the entire edifice of knowledge in medicine was built upon a falsehood."
  - Gøtzsche (BMJ EBM), 2019: "The Cochrane HPV vaccine review was incomplete and ignored important evidence of bias" (only 26 out of 46 trials included)

Section 1

5 minutes of something practical

Implementing meta-analyses

1. R is a good tool for meta-analyses. Reasons:
   - Meta-analyses are not computationally intensive
   - A lot of data processing, visualisation, need for a specialised ecosystem (do not code your own solutions!)
   - Big focus on reporting to non-technical audiences
2. General R packages: `meta`, `metafor` (frequentist), `baggr` (Bayesian)
3. Each of them has good tutorials (look online or in `vignettes()` in R)

Quick worked example in R (frequentist)

```r
library(metafor)

data(schools, package = "baggr")

# RE meta-analysis model
res <- rma(yi = tau, sei = se, data = schools, method = "REML"

res # Inspect result
forest(res, xlab = "SAT improvement") # Forest plot
funnel(res) # Funnel plot (not very useful with K=8)
```

Section 2

A bit more formal look

Three canonical models

Fixed-Effect Meta-Analysis: assumes a single true effect size across all studies

Random-Effects Meta-Analysis: accounts for both within-study and between-study variability in effect sizes.

Meta-regression: adds study-level explanatory variables for effect sizes (e.g. "effect sizes depend linearly on take-up of intervention")

Inference problem

Sampling model: $b_i \sim \mathcal{N}(\theta_i, se_i^2)$

Between-study model: $\theta_i \sim \mathcal{N}(\mu, \sigma^2)$

Let us infer μ and σ.

$$b_i \mid \mu, \sigma^2 \sim \mathcal{N}(\mu, se_i^2 + \sigma^2)$$

Fixed-effect model (inverse variance weights):

$$\sigma^2 = 0 \quad \Rightarrow \quad \hat{\mu} = \frac{\sum_i b_i / se_i^2}{\sum_i 1/se_i^2}$$

Random-effects model (estimate $(\hat{\sigma}, \hat{\mu})$ jointly!):

$$\sigma^2 > 0 \quad \Rightarrow \quad \hat{\mu} = \frac{\sum_i b_i / (se_i^2 + \hat{\sigma}^2)}{\sum_i 1/(se_i^2 + \hat{\sigma}^2)}$$

Solving the inference problem

Frequentist:

- FE is easy (previous slide)—unless the studies don't behave nicely
- RE can be closed form (Der Simonian and Laird) or restricted maximum likelihood
- Many different estimators to choose from, this can matter a lot in edge cases

Bayesian:

- Set priors $p(\mu)$ and $p(\sigma)$
- Then $p(\mu, \sigma|\text{data}) \propto p(\mu)p(\sigma)p(\text{data}|\mu, \sigma)$
- This can matter even more!

Back to the example in R: Bayesian

```r
library(baggr)

data(schools, package = "baggr")

# RE meta-analysis model
bg <- baggr(schools)

bg # Inspect result
plot(bg) + xlab("SAT improvement") # Forest plot
funnel(bg) # Funnel plot (not very useful with K=8)
```

Exchangeability

Meta-analysis assumes studies are *exchangeable.*

- Formally: joint distribution of random variables is invariable to permutations.
- Or: before seeing the data, we have no principled reason to expect one study's true effect to be systematically larger or smaller than another
- Essential assumption of meta-analytic model. Often debatable!
- Can be conditional on some covariates (next slide)

Meta-regression

Explain part of heterogeneity among studies by using regression on study-level covariates.

$$\theta_i \mid \mu, \beta, \sigma^2 \sim \mathcal{N}(\mu + x_i^\top \beta,\ \sigma^2)$$

It's a simple technique. Applicable to both FE and RE models. Can even do it with individual-level data.

Individual-level data

Also referred to as "IPD" in medicine. Let j = 1,. . . ,n_i index individuals within study i.

Outcome model:

$$y_{ij} \mid \alpha_i, \theta_i, z_{ij} \sim \mathcal{N}(\alpha_i + \theta_i\, t_{ij} + z_{ij}^\top \gamma,\ \sigma_y^2)$$

Study-level model for treatment effects:

$$\theta_i \mid \mu, \beta, x_i, \sigma^2 \sim \mathcal{N}(\mu + x_i^\top \beta,\ \sigma^2)$$

Study-level model for intercepts:

$$\alpha_i \sim \mathcal{N}(\alpha_0,\ \sigma_\alpha^2)$$

Hierarchial modelling

Meta-analysis is an example of a hierarchical model

- $y_{ij} = \alpha + \beta\theta_i + \epsilon_{ij}$, with $\epsilon_{ij} \sim \mathcal{N}(0, \tau^2)$
- $\theta_i \sim \mathcal{N}(\mu, \sigma^2)$
- also known as multilevel models, mixed models

You can complicate this:

- many levels: several effects per study, several studies per publication
- covariates at each level (meta-regression!)
- corrections for re-using the sample
- multi-dimensional outcomes

Predicting effect in a new implementation

Estimates of μ and σ are historical means

Typically you care about out-of-sample observations: "what will be the effect of intervention in an yet-unobserved study?"

- Uncertainty about μ
- Heterogeneity σ
- Uncertainty about σ

BLUE/BLUP estimation in frequentist More intuitive in Bayesian inference: for posterior vectors of draws from μ and σ, `mu_post`, `sigma_post`, do:

```
rnorm(N, mu_post, sigma_post)
```

That's it! You can & should do it even for a single study (van Zwet, Gelman, and Więcek 2025)

Reporting meta-analysis results

Forest plot: show effect sizes and confidence intervals (CIs) from each study included in a meta-analysis alongside pooled estimate

(revisit our example)

Publication bias

- People may choose to not publish non-significant results
- We can inspect funnel of effect sizes vs standard errors
  - Small non-significant studies missing? Parts of the funnel empty?
  - Large proportion of studies outside of funnel?
  - You can test for asymmetry (or relationship between SE and effects)
- Funnel shapes diagnose many issues, not publication bias
- You can estimate relative probability of publication (Hedges' selection model, 1992)
  - Can even correct the estimate for this probability

Why be Bayesian?

- We know something about plausible effect sizes or heterogeneity—use it as a prior
  - For example based on non-experimental data
  - or effect sizes in similar interventions
- Frequentist meta-analyses can underestimate heterogeneity (often snap to zero)
- Easier to build complicated models with Bayesian inference
- Posterior predictive distributions and cost-effectiveness analyses

Conclusion to this intro

Terms we introduced:

- meta-analysis,
- effect size,
- generalisability,
- heterogeneity,
- fixed effects,
- random effects,
- meta-regression,
- forest plot

Questions?

Section 3

Scrap material

Another example

```r
library(meta)
# Import data from CSV file
data <- read.csv("meta_data.csv")
# Compute effect sizes and standard errors
effect_size <- log(data$odds_ratio)
# Typical way of approximating SE(log(OR)), we will discuss this later
se <- sqrt(1/data$a + 1/data$b + 1/data$c + 1/data$d)
# Create meta-analysis object
meta_result <- meta::metagen(effect_size, se, studlab = data$study_name)
# Forest plot
meta::forest(meta_result, xlim = c(-5, 3), xlab = "Log Odds Ratio",
             atransf = exp, cex = 0.8,
             print.tau2 = FALSE,
             comb.random = TRUE,
             comb.fixed = TRUE)
# Meta-regression (if applicable)
meta::metareg(meta_result, ~covariate1 + covariate2)
# Funnel plot (if applicable)
meta::funnel(meta_result)
```